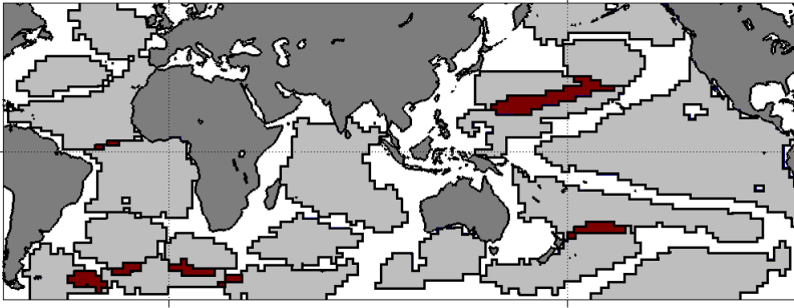


L8: δ -MAPS – Domain Homogeneity Constraint



In the following we simplify the presentation of the delta-MAPS method – the reader can find additional information in the following research paper:

[δ-MAPS: From spatio-temporal data to a weighted and lagged network between functional domains](https://link.springer.com/article/10.1007/s41109-018-0078-z) 
(<https://link.springer.com/article/10.1007/s41109-018-0078-z>)

The input is a timeseries $x_i(t)$ for each grid point i .

The similarity between two grid points i and j is quantified using Pearson's cross-correlation $r_{i,j}$.

More advanced correlation metrics could be used instead (such as mutual information). Additionally, the correlation could be calculated at different lags.

A domain $A(s)$ with seed s (a *specific grid point*) is the largest possible spatially contiguous set of points including s such that their average pairwise correlation $\hat{r}(A)$ is greater than the parameter of the method δ .

The objective of Delta-MAPS is to identify the minimum number of such maximal “domains” in the given data.

For example, the visualization shows the detected domains when Delta-MAPS is applied to Sea Surface Temperature (SST) monthly data (*HadISST*) for the time period of about 50 years (*after removing the effects of seasonality*). Each spatially contiguous region is a domain. You can think of a domain as a network community in which the nodes are grid points in space, and the edge weight between two nodes is equal to the cross-correlation of their activity time series.

Note that there are 18 such domains in this data. The red regions correspond to spatial overlaps between adjacent domains – Delta-MAPS allows for overlapping communities.

Another feature of the method is that not every node needs to belong to a community – as we see there are many grid points on the oceanic surface that do not belong to any domain.

Food for Thought

Can you think of different ways to state (not solve) the optimization problem that Delta-MAPS focuses on? What could be another way to define a spatially contiguous domain (instead of using a constraint on the average pairwise correlation)?